

Scenario 3: Architectural Roadmap and Key Deliverables

The infrastructure engineering team must prioritize the development and validation of these security and performance features:

- Implement OAuth2 and OpenID Connect for secure machine-to-machine identity validation.
- Deploy highly available Kubernetes pods across multiple distinct geographic cloud regions.
- Integrate Model Context Protocol (MCP) servers to allow structural tool-calling capabilities.
- Conduct extensive zero-trust network packet analysis under simulated vector database query loads.

Failure to deliver these core components before the semester audit will result in a deployment hold.